Seewoo Lee

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Academic Employment

École Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland
Collaborateur scientifique (Postdoc) — 2026 – present

- Mentor: Maryna Viazovska

Education

University of California Berkeley — Berkeley, US
Ph.D. in Mathematics — 2018 – 2026

- Thesis: *Positive Quasimodular Forms and Linear Programming Bounds*
- Advisor: Sug Woo Shin
- On leave for military service (2019 Fall - 2022 Summer)

Pohang University of Science and Technology (POSTECH) — Pohang, South Korea
M.S. in Mathematics — 2017 – 2018

- Thesis: *Maass wave forms, quantum modular forms and Hecke operators*
- Advisor: YoungJu Choie

Pohang University of Science and Technology (POSTECH) — Pohang, South Korea
B.S. in Mathematics — 2013 – 2017

- *Summa Cum Laude* with top honours in mathematics
- Honor's thesis: *Quantum modular forms and Hecke operators*

Work Experiences

Axiom Math — Palo Alto, US
Internship — 2026.01 – 2026.04

- Machine Learning & AI for Mathematical Discovery and Reasoning

CryptoLab — Seoul, South Korea
Research Engineer — 2021.05 – 2022.07

- Research on Homomorphic Encryption and application in Machine Learning

Riiid! — Seoul, South Korea
Research Scientist — 2019.07 – 2021.05

- Research on Knowlege Tracing, Score Prediction, Student Dropout Prediction, Item Recommendation

Research Interests

- Computational Number Theory, Automorphic Forms and Representations, Relative Langlands Program
- AI for Mathematics, Formalization of Mathematics, Homomorphic Encryption

Publications

- Math
  1. S. Hariharan, C. Birkbeck, **S. Lee**, G. Ma, B. Mehta, A. Poiroux, M. Viazovska, *Progress in Formalizing Sphere Packing in Dimension 8*, To appear in Proceedings of the International Congress on Mathematical Software 2026, arXiv:2604.23468
  2. **S. Lee**, *Shanks bias in function fields*, To appear in J. Théor. Nombres Bordeaux, arXiv:2509.16142
  3. **S. Lee**, *Algebraic proof of modular form inequalities for optimal sphere packings*. To appear in Algebra Number Theory, arXiv:2406.14659
  4. G. Bates, R. Jesubalan, **S. Lee**, J. Lu, H. Shim, *Powerful Fibonacci polynomials over finite fields*, Finite Fields Appl. 117 (2027)
  5. E. Chen, K. Lau, **S. Lee**, K. Ono, J. Zhang, *Almost all primes are partially regular*, Arch. Math. (2026)
  6. D. Angdinata, E. Chen, **S. Lee**, K. Ono, J. Zhang, *ABC implies that Ramanujan's tau function misses almost all primes*, Indag. Math. (2026)
  7. K. Lee, **S. Lee**, *Machines Learn Number Fields, But How? The Case of Galois Groups*, Res. Math. Sci. 13, 60 (2026)
  8. J. Baek, **S. Lee**, *An equilateral triangle of side $> n$ cannot be covered by $n^2 + 1$ unit equilateral triangles homothetic to it*, Amer. Math. Monthly, 1-9 (2024)
  9. D. Choi, **S. Lee**, *Non-archimedean Sendov's conjecture*, p-Adic Numbers Ultrametric Anal. Appl. 14, 77-80 (2022)
  10. **S. Lee**, *Maass wave forms, Quantum Modular Forms and Hecke Operators*, Res. Math. Sci. 6, 7 (2018), Modular Forms are Everywhere: Celebration of Don Zagier's 65th Birthday
  11. **S. Lee**, *Quantum Modular Forms and Hecke Operators*, Res. Number Theory 4, 18 (2018)
  12. Y. Chen, R. Chernov, M. Flores, M. F. Bourque, **S. Lee**, B. Yang, *Toy Teichmüller spaces of real dimension 2: the pentagon and the punctured triangle*, Geom. Dedicata 197 (2018), 193-227
- Others
  1. F. Lin, K. Nagel, **S. Lee**, J. Jiang, G. Yang, P. Chang, S. Li, N. Sheu, *An Analysis of Silk Density in Spider Webs*, R. Soc. Open Sci. 12: 250455 (2025)
  2. **S. Lee**, G. Lee, J. Kim, J. Shin, M. Lee, *HETAL: Efficient Privacy-preserving Transfer Learning with Homomorphic Encryption*, International Conference on Machine Learning. 2023 (Oral, 155/6538)
  3. **S. Lee**, J. Kim, *Revisiting the Convergence Theorem for Competitive Bidding in Common Value Actions*, Econ. Theory Bull. 10, 293-302 (2022)
  4. **S. Lee**, K. Kim, J. Shin, J. Park, *Tracing Knowledge for Tracing Dropouts: Multi-Task Training for Study Session Dropout Prediction*, Educational Data Mining. 2021
  5. M. Kim, Y. Shim, **S. Lee**, H. Loh, J. Park, *Behavioral Testing of Deep Knowledge Tracing Models*, Educational Data Mining 2021

6. H. Loh, D. Shin, **S. Lee**, J. Baek, C. Hwang, Y. Lee, Y. Cha, S. Kwon, J. Park and Y. Choi, *Recommendation for Effective Standardized Exam Preparation*, LAK21: 11th International Learning Analytics and Knowledge Conference. 2021
7. D. Shin, Y. Shim, H. Yu, **S. Lee**, B. Kim, Y. Choi, *SAINT+: Integrating Temporal Features for EdNet Correctness Prediction*, LAK21: 11th International Learning Analytics and Knowledge Conference. 2021
8. Y. Choi, Y. Lee, D. Shin, J. Cho, S. Park, **S. Lee**, J. Baek, C. Bae, B. Kim, J. Heo, *EdNet: A Large-Scale Hierarchical Dataset in Education*, International Conference on Artificial Intelligence in Education (2021), 69-73
9. J. Kim, **S. Lee**, *Joint Liability and Stochastic Shapley Value*, Int. Rev. Law Econ. 60 (2019), 1-8

Preprints

1. **S. Lee**, *Positive quasimodular forms and the sign uncertainty principle*, arXiv:2608.15415, submitted
2. B. Banwait, X. Huang, K. Lee, **S. Lee**, T. Oliver, A. Podznyakov, *Decision trees, Frobenius traces, and Weierstrass coefficients of elliptic curves* arXiv:2607.24251, submitted
3. K. Lau, **S. Lee**, K. Ono, *Formalized q-series: The Rogers-Ramanujan Identities and Beyond*, arXiv:2607.01544, submitted
4. **S. Lee**, B. Hwang, H. Lim, J. Hyun, I. Choi, Y. Park, J. Baek, H. Hong, K. Lee, J. Heo, H. Baik, C. Lee, K. Lee, *Lean-GAP: A Dataset of Formalized Graduate Algebra Problems* arXiv:2606.02588, submitted
5. G. Bates, R. Jesubalan, **S. Lee**, J. Lu, H. Shim, *Ties in function field prime race*, arXiv:2603.21005, submitted
6. **S. Lee**, *Inequalities involving polynomials and Quasimodular Forms*, arXiv:2602.10536, submitted
7. E. Chen, K. Lau, **S. Lee**, K. Ono, J. Zhang, *Dead ends in square-free digit walks*, arXiv:2602.05095
8. J. Getz, A. G. Terradillos, F. Hosseinijafari, B. Hu, **S. Lee**, A. Slipper, M.-H. Tomé, H. Yao, A. Zhao, *Modulation groups*, arXiv:2510.23932, submitted
9. J. Baek, **S. Lee**, *Formalizing Mason–Stothers Theorem and its Corollaries in Lean 4*. arXiv:2408.15180
10. **S. Lee**, Y. Choi, J. Park, B. Kim, J. Shin, *Consistency and Monotonicity Regularization for Neural Knowledge Tracing*, arXiv:2105.00607
11. Y. Choi, Y. Lee, J. Cho, J. Baek, D. Shin, H. Yu, Y. Shim, **S. Lee**, J. Shin, C. Bae, B. Kim, J. Heo, *Assessment Modeling: Fundamental Pre-training Tasks for Interactive Educational Systems*, arXiv:2022.05505

Awards, Grants & Honours

Nikki Kose Memorial Teaching Prize, Dept. of Mathematics, UC Berkeley . . . . 2026 Spring
Department of Mathematics Summer Grant, UC Berkeley . . . . 2024 Summer
Outstanding Graduate Student Instructor Award, UC Berkeley . . . . 2024 Spring
Graduate Student Researcher, UC Berkeley . . . . 2023 Spring, Summer
Kwanjeong Educational Foundation Scholarship, KEF 2017–2018
Excellency Award (Top Honours), Dept. of Mathematics, POSTECH 2017
POSTECH Outstanding Talent Development Scholarship, POSTECH 2013–2016
National Science and Technology Scholarship, KOSAF 2013–2016
Silver medals, Undergraduate Mathematical Competition, KMS 2013, 2015, 2016
31st place, ACM-ICPC Daejeon Regional, ACM 2015
Grand prize, POSTECH Programming Contest, Dept. of Computer Science, POSTECH 2015
Honorable mention, Korean Olympiad of Informatics, NIA 2012

Teaching Experience

T.A. Lausanne
EPFL 2026 – Present

- (2026 Fall) Algèbre I - structures fondamentales

Graduate Student Instructor (T.A.) Berkeley
UC Berkeley 2019 – 2026

- (2026 Spring) Multivariable Calculus
- (2025 Fall) Introduction to Abstract Algebra
- (2025 Spring) Cryptography
- (2025 Spring) Introduction to Mathematical Logic
- (2024 Fall) Abstract Linear Algebra
- (2024 Spring) Methods of Mathematics: Calculus, Statistics, and Combinatorics
- (2023 Fall) Methods of Mathematics: Calculus, Statistics, and Combinatorics
- (2022 Fall) Multivariable Calculus
- (2019 Spring) Methods of Mathematics: Calculus, Statistics, and Combinatorics

Grader & T.A. Pohang
POSTECH 2016 – 2018

- (2018 Spring) Differential Manifolds and Lie groups (Graduate course)
- (2017 Fall) Modern Algebra II
- (2017 Spring) Calculus
- (2016 Fall) Applied Linear Algebra (Undergraduate T.A.)

Tutoring Pohang
POSTECH 2014 – 2015

- (2015 Spring) Calculus
- (2015 Spring) Modern Algebra I
- (2014 Fall) Analysis II
- (2014 Spring) Analysis I

Outreach

KIAS Thematic Program on AI and Mathematics Seoul
Coordinator for week 2: Number Theory 2026 Summer

- Organized a week-long program on Number Theory and AI

KIAS Winter School on Mathematics and AI Jeongseon
Team Leader 2025 Winter

- Use machine learning to study Artin representations and abelian varieties over finite fields
- Members: Dohoon Choi, Keunyoung Jeong, Jaehak Yi, Minseo Shin, Euntaek Lee, Taeyoung Kim

Berkeley Math REU Berkeley
Mentor 2025 Summer

- Diophantine equations on Fibonacci polynomials and ties in Chebyshev bias over finite fields
- Mentees: Graeme Bates, Ryan Jesubalan, Jane Lu, Hyewon Shim

Directed Reading Program Berkeley
Mentor 2023-present

- (2025 Fall) Group cohomology (Graeme Bates, Jane Lu)
- (2025 Spring, 2025 Fall) Modular forms (Dongho Kim)
- (2023 Fall) Elliptic curves (Jacob Martin)
- (2023 Spring) p-adic numbers (Lucas Xie)

POSTECH Potential Development Camp for High School Students POSTECH
Mentor 2015 Winter

- Mentoring high school students for college-level mathematics

Talks

- Research Talks
  - Thematic Program on AI and Mathematics, KIAS, Seoul, July 2026
    Formalized q-series
  - BRL-AGSTA One-day Workshop, DGIST, Daegu, July 2026
    Machine Learning Number Theory
  - Mathematics x AI: Challenges and Opportunities, KIAS, Seoul, July 2026
    Machine Learning Number Theory
  - Workshop on Algebraic Geometry and Machine Learning, ICARM, Pittsburgh, June 2026
    Machine Learning Number Theory
  - Graduate School of AI for Math, KAIST, Daejeon, June 2026
    Positive Quasimodular Forms and Linear Programming Bounds
  - Mathematics Colloquium, Carnegie Mellon University, Pittsburgh, April 2026
    Positive Quasimodular Forms and Linear Programming Bounds
  - Aarhus Mathamtics & AI Workshop, Aarhus Institute of Advanced Studies, Aarhus, January 2026
    How Machines Learn Galois Groups
  - Lean Together 2026, Online, January 2026
    Progress on the Sphere Packing Project (with Sidarth Hariharan and Chris Birkbeck)

- June E Huh Center for Mathematical Challenges, KIAS, Seoul, January 2026
  Positive Quasimodular Forms and Linear Programming Bounds
- Number Theory Seminar, University of Wisconsin-Madison, Wisconsin, December 2025
  How Machines Learn Galois Groups
- EPFL, Lausanne, November 2025
  Sphere Packing, Sign Uncertainty Principle, Universal Optimality, and Positive Quasimodular Forms
- Aarhus University, Aarhus, November 2025
  Sphere Packing, Sign Uncertainty Principle, Universal Optimality, and Positive Quasimodular Forms
- Special Seminar, Carnegie Mellon University, Pittsburgh, November 2025
  How Machines Learn Galois Groups
- International Seminar on Automorphic Forms, Online, April 2025
  Algebraic proof of modular form inequalities for optimal sphere packings
- RTG seminar, UC Berkeley, Berkeley, February 2025
  Algebraic proof of modular form inequalities for optimal sphere packings
- Algebra Discrete Math seminar, UC Davis, Davis, January 2025
  Algebraic proof of modular form inequalities for optimal sphere packings
- 6th EU/US Workshop on Automorphic Forms and Related Topics, Luminy, September 2024
  Algebraic proof of modular form inequalities for optimal sphere packings
- POSTECH Number Theory Seminar, POSTECH, Pohang, May 2024
  Algebraic proof of modular form inequalities for optimal sphere packings
- Student Number Theory Seminar, UC Berkeley, Berkeley, April 2024
  Algebraic proof of Viazovska's inequalities
- School of Mathematics, KIAS, Seoul, December 2023
  A new proof of Viazovska's modular form inequality and beyond
- International Conference on Machine Learning, Hawaii, US, July 2023
  HETAL: Efficient Privacy-preserving Transfer Learning with Homomorphic Encryption
- Center for Artificial Intelligence and Natural Sciences, KIAS, Seoul, June 2023
  HETAL: Efficient Privacy-preserving Transfer Learning with Homomorphic Encryption
- School of Computing, KAIST, Daejeon, June 2023
  HETAL: Efficient Privacy-preserving Transfer Learning with Homomorphic Encryption
- 1st FHE.org workshop, Trondheim, May 2022
  Encrypted Multinomial Logistic Regression Training with Softmax Approximation
- Workshop for Young Mathematicians in Korea, Online, January 2022
  Hitchhiker's guide to non-archimedean world
- Graduate student seminar, Sogang University, Seoul, July 2018
  Maass wave forms, quantum modular forms and Hecke operators
- Sungkyunkwan University, Seoul, June 2018
  Maass wave forms, quantum modular forms and Hecke operators
- NCTS-POSTECH Number Theory Workshop, NTU, Taiwan, December 2017
  Quantum modular forms and Hecke operators

• Expository Talks

- Korea University, Seoul, June 2026
  Mathematics, AI, and Formalization

– Student Number Theory Seminar, Berkeley, March 2026
  Mathematics, AI, and Formalization
– Harmonic Analysis Seminar, University of California Irvine, February 2026
  Mathematics, AI, and Formalization
– Student Number Theory Seminar, Berkeley, August 2025
  How Do Automorphic Forms and Elliptic Curves Fly? (Survey on Murmuration)
– Bruhat-Tits building seminar, Berkeley, February 2025
  Bruhat-Tits buildings for split groups / Moy-Prasad filtration and Local Langlands correspondence
– Berkeley–Stanford Number Theory Learning Seminar, Berkeley, December 2024.
  Proof of irrationality of $L(2, \chi_{-3})$ and product of log values
– Student Number Theory Seminar, Berkeley, October 2024
  Modular forms on G_2
– Geometric class field theory learning seminar, Berkeley, Sep 2024
  Singular algebraic curves and de-normalization
– Student Number Theory Seminar, Berkeley, March 2024
  Linear Programming Beyond Sphere Packing
– Orbit methods and automorphic forms learning seminar, Berkeley, Oct 2023
  Gan–Gross–Prasad conjectures
– Student Number Theory Seminar, Berkeley, Nov 2022
  Shimura correspondence and Waldspurger's formula
– Instructional Workshop on Class Field Theory, KIAS, Seoul, January 2018
  Proof of the main theorem of local class field theory

Languages

- Korean (native), English (fluent)
- Python (PyTorch, Numpy, Pandas), C/C++, LaTeX, SageMath, Lean, MATLAB, Haskell

Miscelleneous (click the icons)

- Working as a reviewer for Mathematical Reviews (2022∼)
- GitHub blog on various topics
- Math Stackexchange & Math Overflow
- Speedcuber
- DJ (Techno, Trance, House)

(Last updated: September 2, 2026)